



# Challenge Guidelines

## 1. Main Objective

The objective of the OpenFOAM Hackathon Challenge is to **quantitatively assess how different sustainable mobility scenarios affect air quality in Guimarães**, using the provided reference case as the baseline for comparison.

The assessment must focus on pollutant concentration levels at the following selected sensitive locations, whose location is provided in file "ROI/ROI.obj":

- Martins Sarmiento High School
- Francisco de Holanda High School
- Santos Simões High School
- Public Hospital

Each team should compare the results obtained for the proposed scenarios against the reference case and provide a clear quantitative evaluation of the expected air-quality improvement.

## 2. The Reference Case

The following folder contains the data required to set up and run the reference case:

<https://nextcloud.crheo.dep.uminho.pt/index.php/s/HXMtcaLNnTzx47>

The available data includes:

- **Building and terrain surfaces** used to construct the computational domain. Teams are free to use alternative datasets or improve the provided geometry if they believe this will lead to more accurate or robust results (***note that the geometry was changed from the original one to accomplish the best practice guidelines presented in Track 1***).
- **High-canopy surfaces** representing vegetation and other relevant canopy features across the study area.
- **Road segments** mapped onto the provided terrain model. If a different terrain representation is adopted, remapping the road network should require only limited effort.
- **Emission data** for two pollutants (**CO** and **NOx**), provided at **one-hour intervals over a full day**.
- **Wind data**, also provided at **one-hour intervals over a full day**, to be used as atmospheric boundary conditions.



The provided dataset for the Reference Case should be considered the baseline information required to address the challenge. Teams are encouraged to improve the modelling approach, geometry representation, meshing strategy, workflows, and analysis tools whenever appropriate, provided that the final results remain comparable with the reference case and the challenge objectives

### 3. Mobility Scenarios to Assess

Each team is expected to evaluate the following four scenarios:

- **Reference Case**

Data provided in Section 2

- **Scenario 1 – 20% Reduction of Gas-Based Vehicle Traffic**

Reduction of **20% of traffic from gas-based vehicles**, assuming replacement by electric vehicles.

- **Scenario 2 – 40% Reduction of Gas-Based Vehicle Traffic**

Reduction of **40% of traffic from gas-based vehicles**, assuming replacement by electric vehicles.

- **Scenario 3 – Metro Bus Implementation in Guimarães**

Assessment of the impact of introducing a **Metro Bus Between Guimarães and Braga**, assumed to reduce traffic on the N101 by approximately **50%**. For this scenario, teams should apply the **traffic reduction** in the street segments indicated in file “road\_ids\_reduction.txt”

### 4. Expected Outputs

Each team should provide:

- OpenFOAM cases and tools used to run the simulations;
- post-processing tools or scripts used to quantify air quality;
- quantitative comparison between the reference case and the three mobility scenarios;
- assessment of the influence of the computational domain;
- clear visualizations of pollutant concentration fields;
- Proposals for the continuation of the work.



## 5. Assessment Criteria

The final evaluation will consider the following criteria:

| Criterion                                                                | Weight |
|--------------------------------------------------------------------------|--------|
| Correct implementation of the mobility scenarios                         | 25%    |
| Quantitative assessment of air quality at the predefined locations       | 20%    |
| Quality and robustness of the OpenFOAM workflow and tools                | 30%    |
| Quality of post-processing, visualization, and interpretation of results | 10%    |
| Clarity of the final presentation and conclusions                        | 15%    |

## 6. Deliverables

Each team is expected to submit its final deliverables no later than **July 2<sup>nd</sup> at 2PM (Portugal Time Zone)**.

The submitted package should include:

- OpenFOAM cases and tools developed during the Hackathon;
- Any additional scripts, utilities, or workflows created by the team;
- Post-processing tools used to quantify air quality;
- A concise technical report (PDF) summarizing:
  - The methodology followed;
  - The mobility scenarios assessed;
  - The quantitative impact of each mobility scenario on air quality at the target locations;
  - The main conclusions and recommendations for future work.

All submitted materials must be sufficiently documented to allow the jury to understand the work performed and reproduce the main results if necessary.



## 7. Final Presentations

On the **afternoon of July 3<sup>rd</sup>**, during OFW21, each team will present its work to the Hackathon participants, OFW21 attendees, and the evaluation jury.

Each team will be allocated:

- 15 minutes for the presentation;
- 5 minutes for questions and discussion.

The presentation should clearly describe:

1. The team organization and contribution of the different tracks;
2. The tools and workflows developed;
3. The simulation strategy adopted;
4. The assessment of the four mobility scenarios;
5. The main findings regarding air quality improvements;
6. Recommendations for future work.

The quality of the final presentation will be considered as part of the overall evaluation.

## 8. Awards

A total prize of €2000 (two thousand euros) will be awarded to the participating teams.

The jury reserves the right to decide how the prize money will be distributed among the teams, based on the overall quality of the submitted work, the final presentation, and the evaluation criteria defined in these guidelines.

The jury may choose to:

- Award the full amount to a single team;
- Distribute the prize among multiple teams;
- Create specific awards recognizing particular achievements (e.g., best technical solution, best automation workflow, best HPC/ML contribution, best presentation).

The jury's decision will be final.